

BST Physics Curriculum Vol 10 Chapter 5 — Fourier Analysis + Distributions v0.4 (refilled per Cal #104)

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Vol 10 Chapter 5 — Fourier Analysis + Distributions

Chapter motivation

Standard graduate Fourier analysis: Fourier series for periodic functions ($\sum c_n e^{inx}$); Fourier transform on $\mathbb{R}^n$ ($Ff = \int f(x) e^{-ik \cdot x} dx$); Plancherel theorem (Fourier transform unitary on L^2); Schwartz distributions + delta function $\delta(x)$; Sobolev spaces. Standard texts: Stein-Shakarchi + Hörmander + Reed-Simon.

BST cross-link: Wallach 1976 K-type decomposition of Bergman $H^2(D_{IV}^5)$ (Vol 1 Ch 2 + Vol 11 Ch 3) IS the substrate-Fourier framework. K-types $V_{-}(p,q)$ under $K = SO(5) \times SO(2)$ replace standard Fourier modes $e^{ik \cdot x}$; Casimir eigenvalue spectrum (Vol 1 Ch 5) replaces standard frequency spectrum. Substrate-Fourier decomposition fundamental to all BST physics.

Section 5.0b — Reader-grade 3-level pedagogy

Level 1: Standard Fourier transform (decomposition into frequency modes) + delta distributions; BST cross-link: Wallach K-type decomposition of Bergman $H^2(D_{IV}^5)$ IS substrate-Fourier — substrate “frequency modes” are Wallach K-types $V_{-}(p,q)$ with Casimir eigenvalue spectrum.

Level 2 (graduate-physicist): Standard Fourier theory: periodic functions decomposed as $\sum_n c_n e^{inx}$; non-periodic functions decomposed via $Ff = \int f(x) e^{-ik \cdot x} dx$ (Fourier transform); Plancherel theorem $\|F[f]\|^2 = (2\pi)^n \|f\|^2$ makes F unitary on $L^2(\mathbb{R}^n)$; delta function $\delta(x) = (1/(2\pi)^n) \int e^{ik \cdot x} dk$ (distribution-theoretic identity). Schwartz distributions extend functions to “generalized functions” including δ , principal value $1/x$, etc.; Sobolev spaces H^s control derivative regularity. BST substrate cross-link: per Wallach 1976 (Vol 11 Ch 3), Bergman space $H^2(D_{IV}^5)$ decomposes under $K = SO(5) \times SO(2)$ into countably-infinite direct sum $H^2 = \oplus V_{-}(p,q)$ of irreducible K-types $V_{-}(p,q)$

labeled by integer pairs (p, q) admissibility. K-types $V_{-}(p, q)$ play role of standard Fourier modes $e^{ik \cdot x}$; Casimir eigenvalue $\text{Cas}(p, q)$ of K-type $V_{-}(p, q)$ plays role of standard frequency $|k|^2$. Substrate-Fourier decomposition replaces standard Fourier in BST physics. Standard Fourier recovered at conformal boundary of D_{IV}^5 (Vol 0 Ch 4 §4.6 + Vol 5 Ch 1 pedagogical bridge): $SO_0(5, 2) \rightarrow SO_0(3, 1) \times SO(2)$ Lorentz embedding gives 4D Lorentz harmonic decomposition = standard 4-momentum Fourier. Substrate delta function = Bergman reproducing kernel evaluation $K_B(z, \bar{w}) \rightarrow \delta(z - w)$ at appropriate boundary limit. Substrate Plancherel: orthonormality of Wallach K-types per Wallach 1976 + Faraut-Koranyi 1994 $c_{FK} \cdot \pi^{(9/2)} = 225$ EXACT normalization (T2442).

Level 3 (5th-grader accessible): Fourier analysis decomposes signals into frequencies (like how a prism decomposes white light into rainbow colors). Standard Fourier transforms decompose functions into e^{ikx} “frequency modes”. BST identifies Wallach K-types (representations of substrate isotropy $SO(5) \times SO(2)$) as the substrate-natural “frequency modes” — substrate physics IS K-type decomposition on Bergman $H^2(D_{IV}^5)$.

Section 5.1 — Standard Fourier Series + Transform

Periodic f : $\sum_n c_n e^{inx}$ with $c_n = (1/2\pi) \int f(x) e^{-inx} dx$.

Non-periodic on $\mathbb{R}^n$: $Ff = \int f(x) e^{-ik \cdot x} dx$.

Plancherel: $\|F[f]\|^2 = (2\pi)^n \|f\|^2$ (Fourier transform unitary on L^2).

Section 5.2 — Distributions + Delta Function

Schwartz distributions extend functions to generalized: $\delta(x)$ defined by $\int \delta(x) \varphi(x) dx = \varphi(0)$ for test functions φ .

Identity: $\delta(x) = (1/(2\pi)^n) \int e^{ik \cdot x} dk$.

Sobolev spaces H^s define derivative-regularity scale.

Section 5.3 — Wallach K-Type Decomposition (Vol 11 Ch 3)

Per Wallach 1976: Bergman $H^2(D_{IV}^5) = \oplus \{(p, q) \text{ admissible}\} V_{-}(p, q)$ decomposition under $K = SO(5) \times SO(2)$.

K-types $V_{-}(p, q)$ labeled by integer pairs (p, q) satisfying admissibility conditions from Wallach + Faraut-Koranyi 1994.

Section 5.4 — Substrate-Fourier Framework

Wallach K-types $V_{-}(p, q)$ play role of standard Fourier modes $e^{ik \cdot x}$.

Casimir eigenvalue $\text{Cas}(p, q)$ of $V_{-}(p, q)$ plays role of standard frequency $|k|^2$.

Substrate-Fourier decomposition: $f \in \text{Bergman } H^2(D_{IV^5}) = \sum_{\{p,q\}} c_{\{p,q\}} V_{\{p,q\}\text{-component}}$.

Section 5.5 — Standard Fourier Recovery at Conformal Boundary

Per Vol 0 Ch 4 §4.6 + Vol 5 Ch 1 pedagogical bridge:

$SO_0(5,2) \rightarrow SO_0(3,1) \times SO(2)$ Lorentz embedding gives 4D Lorentz harmonic decomposition = standard 4-momentum Fourier.

Substrate delta = Bergman kernel $K_B(z, \bar{w})$ at appropriate boundary limit $\rightarrow$ standard $\delta(z - w)$.

Plancherel: substrate K-type orthonormality + Faraut-Koranyi $c_{FK} \cdot \pi^{(9/2)} = 225 \text{ EXACT (T2442)}$ gives substrate Plancherel; standard Plancherel recovered at boundary.

Section 5.6 — Honest scope + Connection

- Standard Fourier + distributions [?]
- Wallach K-type substrate-Fourier cross-link [?]
- Standard Fourier recovery at conformal boundary
- Open scope: explicit $V_{\{p,q\}} \rightarrow e^{\{ik \cdot x\}}$ reduction at conformal boundary (multi-week)

Connection: - Vol 1 Ch 2 Substrate Hilbert Space (Bergman H^2 + Wallach K-type)
 - Vol 11 Ch 3 Wallach K-Type Representation Theory - Vol 10 Ch 4 PDEs (separation of variables $\rightarrow$ K-type decomposition) - Vol 1 Ch 5 Casimir algebra (Cas(p, q) eigenvalue spectrum)

— Lyra, Vol 10 Ch 5 v0.4 chapter-grade narrative REFILLED per Cal #104, Saturday 2026-05-23 EDT